Airline Passenger Satisfaction

Micro-Project 1

Acquire and Prepare

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Problem Statement

- Passenger satisfaction varies across airline travel experiences.
- I will examine whether longer arrival delays are associated with a lower likelihood of passenger satisfaction.
- Outcome: satisfied versus neutral or dissatisfied. The latter combines two responses.
- This dataset can support an observational study; it cannot establish that delays cause dissatisfaction.

Hypothesis Formulation

- Hypothesis: Passengers with longer arrival delays are less likely to report satisfaction.
- Null hypothesis: Arrival delay has no association with satisfaction.
- Travel class, travel purpose, and service ratings may also relate to satisfaction and should be considered in a later study.
- Micro-Project 1 ends at Prepare. No hypothesis test or predictive model is performed.

Acquire

- Input: airline.csv supplied for this course project.
- 129,880 rows and 25 original columns; 129,880 unique passenger IDs.
- Fields include demographics, travel class, service ratings, delays, and satisfaction.
- Load with pandas.read_csv; retain an unchanged raw DataFrame.
- Original publisher, collection dates, and sampling design are not documented in the supplied file; representativeness is unknown.

Prepare

- Organized names and text; removed exported index; kept passenger ID.
- No exact duplicate rows or duplicate IDs; all 129,880 records retained.
- Arrival delay: 393 missing (0.303%); original values preserved, with a missing flag and separate median-filled column (0 minutes).
- Checked age (7-85), positive distance, nonnegative delays, and ratings (0-5). Retained rating zeros because their meaning is undocumented.
- Satisfied: 56,428 (43.45%); neutral or dissatisfied: 73,452 (56.55%). Hypothesis remains untested.
- Prepared output: 27 columns; pandas categories, NumPy arrays, and documented validation checks.